

Question:

A baby is born after the 32nd gestational week by cesarean delivery. The mother suffered from gestational diabetes; however, she had no other pregnancy-related diseases and was otherwise healthy. The baby has a blood pressure of 100/58 mm Hg, heart rate of 104/min, and oxygen saturation of 88%. The child has tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis. The cyanosis is responding well to initial administration of oxygen. The nasogastric tube was positioned without problems. Which of the following is the most likely diagnosis?

Answer Choices:

- A. Neonatal respiratory distress syndrome (NRDS)
- B. Tracheoesophageal fistula
- C. Pneumonia
- D. Congenital heart anomaly with right-to-left shunt

Reasoning + Answer:

<reasoning process here> The baby was born at 32 weeks gestation, which is considered preterm, and the presence of tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis are classic signs of respiratory distress. The oxygen saturation is 88%, which is low and consistent with respiratory compromise. The fact that the cyanosis is responding to oxygen administration suggests that the issue is not a chronic or structural problem like a congenital heart anomaly, which would typically not improve with oxygen alone.

Neonatal respiratory distress syndrome (NRDS), also known as transient tachypnea of the newborn (TTN) or respiratory distress syndrome, is common in preterm infants due to surfactant deficiency. This is especially relevant in infants born before 34 weeks, and the clinical presentation of tachypnea, retractions, and hypoxia fits NRDS.

Gestational diabetes can increase the risk of respiratory distress in the newborn due to fetal lung immaturity, which further supports the diagnosis of NRDS.

Tracheoesophageal fistula (B) would present with feeding difficulties, vomiting, and airway signs, but not typically with hypoxia and retractions unless it's a complication of a feeding issue. Pneumonia (C) would likely present with fever, increased white blood cells, and signs of infection, which are not mentioned. A congenital heart anomaly with a right-to-left shunt (D) would present with cyanosis that does not improve with oxygen, and would likely have other signs such as polycythemia or heart murmurs.

Thus, the most likely diagnosis is neonatal respiratory distress syndrome (NRDS), which is the most common cause of respiratory distress in preterm infants.

<answer> A </answer>